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MASS TRANSFER ASSEMBLY, DISPLAY PANEL, AND DISPLAY DEVICE

Abstract

A mass transfer assembly includes a temporary substrate and a transfer substrate. The temporary substrate is configured to adhere electrodes of each light-emitting element on a growth substrate, to separate each light-emitting element from the growth substrate, such that a light-emitting body of each light-emitting element is suspended. The transfer substrate is configured to adhere, through adhesion regions, a light-emitting body of each light-emitting element on the temporary substrate, to separate each light-emitting element from the temporary substrate, such that electrodes of each light-emitting element are suspended. One adhesion region adheres a light-emitting body of one light-emitting element. Positions of the adhesion regions on the transfer substrate are in one-to-one correspondence with positions of binding regions on a driving substrate. The transfer substrate is further configured to transfer each light-emitting element to the driving substrate, to bind electrodes of each light-emitting element to a corresponding binding region.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims priority to Chinese Application Patent Serial No. 202410239001.5, filed Mar. 4, 2024, the entire disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD

[0002] This application relates to the field of display technology, and particularly to a mass transfer assembly, a display panel, and a display device having the display panel.

BACKGROUND

[0003] Micro light-emitting diode (LED) display panels have broad application prospects in the display field because of their advantages of good stability, long life, excellent display effect, etc.

[0004] The micro LED display panel generally includes a driving substrate and millions of micro LED elements disposed on one side of the driving substrate. A distance between two adjacent micro LED elements on the driving substrate is mainly determined by a resolution and a size of the micro LED display panel. Micro LED elements are formed on the growth substrate. In order to improve a utilization rate of the growth substrate, a distance between two adjacent micro LED elements on the growth substrate should be as small as possible. However, if the distance between two adjacent micro LED elements on the growth substrate is smaller than the distance between two adjacent micro LED elements on the driving substrate, the micro LED elements on the growth substrate cannot be transferred to the driving substrate at one time, resulting in an increase in workload and cost of transferring of the micro LED elements.

[0005] Therefore, how to transfer the micro LED elements on the growth substrate to the driving substrate at one time to reduce workload and cost of transferring of the micro LED elements has become an urgent problem to be solved by those skilled in the art.

SUMMARY

[0006] The disclosure provides a mass transfer assembly. The mass transfer assembly is used for transferring a plurality of light-emitting elements from a growth substrate to a driving substrate. Each of the plurality of light-emitting elements includes a light-emitting body initially connected with the growth substrate and two electrodes connected with the light-emitting body. The mass transfer assembly includes a temporary substrate and a transfer substrate. The temporary substrate is configured to adhere two electrodes of each of a plurality of light-emitting elements on the growth substrate, to separate the plurality of light-emitting elements from the growth substrate, such that a light-emitting body of each of the plurality of light-emitting elements is suspended. The transfer substrate has a plurality of adhesion regions. The transfer substrate is configured to adhere, through the adhesion regions, a light-emitting body of each of a plurality of light-emitting elements on the temporary substrate, to separate the plurality of light-emitting elements from the temporary substrate, such that two electrodes of each of the plurality of light-emitting elements are suspended. One adhesion region adheres a light-emitting body of one light-emitting element. Positions of the plurality of adhesion regions on the transfer substrate are in one-to-one correspondence with positions of a plurality of binding regions on the driving substrate. The transfer substrate is further configured to transfer a plurality of light-emitting elements on the transfer substrate to one side of the driving substrate, to bind two electrodes of each of the plurality of light-emitting elements to a corresponding binding region of the driving substrate.

[0007] The disclosure further provides a display panel. The display panel includes a driving substrate and a plurality of light-emitting elements disposed on one side of the driving substrate. The plurality of light-emitting elements are transferred by a mass transfer assembly from a growth substrate to the driving substrate. Each of the plurality of light-emitting elements includes a light-emitting body initially connected with the growth substrate and two electrodes connected with the light-emitting body. The mass transfer assembly includes a temporary substrate and a transfer substrate. The temporary substrate is configured to adhere two electrodes of each of a plurality of light-emitting elements on the growth substrate, to separate the plurality of light-emitting elements from the growth substrate, such that a light-emitting body of each of the plurality of light-emitting elements is suspended. The transfer substrate has a plurality of adhesion regions. The transfer substrate is configured to adhere, through the adhesion regions, a light-emitting body of each of a plurality of light-emitting elements on the temporary substrate, to separate the plurality of light-emitting elements from the temporary substrate, such that two electrodes of each of the plurality of light-emitting elements are suspended. One adhesion region adheres a light-emitting body of one light-emitting element. Positions of the plurality of adhesion regions on the transfer substrate are in one-to-one correspondence with positions of a plurality of binding regions on the driving substrate. The transfer substrate is further configured to transfer a plurality of light-emitting elements on the transfer substrate to the side of the driving substrate, to bind two electrodes of each of the plurality of light-emitting elements to a corresponding binding region of the driving substrate.

[0008] The disclosure further provides a display device. The display device includes a housing and a display panel. The display panel is received in the housing. A light-emitting side of the display panel is exposed from the housing. The display panel includes a driving substrate and a plurality of light-emitting elements disposed on one side of the driving substrate. The plurality of light-emitting elements are transferred by a mass transfer assembly from a growth substrate to the driving substrate. Each of the plurality of light-emitting elements includes a light-emitting body initially connected with the growth substrate and two electrodes connected with the light-emitting body. The mass transfer assembly includes a temporary substrate and a transfer substrate. The temporary substrate is configured to adhere two electrodes of each of a plurality of light-emitting elements on the growth substrate, to separate the plurality of light-emitting elements from the growth substrate, such that a light-emitting body of each of the plurality of light-emitting elements is suspended. The transfer substrate has a plurality of adhesion regions. The transfer substrate is configured to adhere, through the adhesion regions, a light-emitting body of each of a plurality of light-emitting elements on the temporary substrate, to separate the plurality of light-emitting elements from the temporary substrate, such that two electrodes of each of the plurality of light-emitting elements are suspended. One adhesion region adheres a light-emitting body of one light-emitting element. Positions of the plurality of adhesion regions on the transfer substrate are in one-to-one correspondence with positions of a plurality of binding regions on the driving substrate. The transfer substrate is further configured to transfer a plurality of light-emitting elements on the transfer substrate to the side of the driving substrate, to bind two electrodes of each of the plurality of light-emitting elements to a corresponding binding region of the driving substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] In order to describe technical solutions of embodiments of the disclosure more clearly, the following will give a brief description of accompanying drawings used for describing the embodiments. Apparently, accompanying drawings described below are merely some embodiments. Those of ordinary skill in the art can also obtain other accompanying drawings based on the accompanying drawings described below without creative efforts.

[0010] FIG. 1 is a schematic diagram illustrating a layer structure of a growth substrate and light-emitting elements provided in a first embodiment of the disclosure.

[0011] FIG. 2 is a schematic structural diagram illustrating a temporary substrate of a mass transfer assembly provided in the first embodiment of the disclosure.

[0012] FIG. 3 is a schematic diagram illustrating adhering of light-emitting elements with the temporary substrate illustrated in FIG. 2.

[0013] FIG. 4 is a schematic diagram illustrating a first structure of a transfer substrate of the mass transfer assembly provided in the first embodiment of the disclosure.

[0014] FIG. 5 is a schematic diagram illustrating adhering of light-emitting elements with the transfer substrate illustrated in FIG. 4.

[0015] FIG. 6 is a schematic diagram illustrating transferring of light-emitting elements to a driving substrate from the transfer substrate illustrated in FIG. 4.

[0016] FIG. 7 is a schematic diagram illustrating a second structure of the transfer substrate of the mass transfer assembly provided in the first embodiment of the disclosure.

[0017] FIG. 8 is a schematic diagram illustrating adhering of light-emitting elements with the transfer substrate illustrated in FIG. 7.

[0018] FIG. 9 is a schematic diagram illustrating transferring of light-emitting elements to the driving substrate from the transfer substrate illustrated in FIG. 7.

[0019] FIG. 10 is a schematic diagram illustrating a layer structure of a display panel provided in a second embodiment of the disclosure.

[0020] FIG. 11 is a schematic diagram illustrating a layer structure of a display device provided in a third embodiment of the disclosure.

DETAILED DESCRIPTION

[0021] In order to facilitate understanding of the disclosure, the disclosure will be described fully below with reference to accompanying drawings. The accompanying drawings illustrate exemplary embodiments of the disclosure. However, the disclosure may be implemented in many different forms and is not limited to the embodiments described herein. Rather, these embodiments are provided to achieve a thorough and complete understanding of disclosed contents of the disclosure.

[0022] The following descriptions of the embodiments are with reference to the attached diagrams to illustrate specific embodiments that the disclosure can be implemented. The serial numbers for the components herein, such as “first”, “second”, etc., are only used to distinguish the objects described and do not have any order or technical meaning. The “connection” and “coupling” mentioned in the disclosure, unless otherwise specified, include direct and indirect connections (couplings). The directional terms mentioned in the disclosure, such as “upper”, “lower”, “front”, “back”, “left”, “right”, “inside”, “outside”, “side”, etc., are only with reference to the directions of the attached drawings. Therefore, the directional terms used are for better and clearer explanation and understanding of the disclosure, rather than indicating or implying that the device or element referred to must have a specific orientation, be constructed and operated in a specific orientation, and therefore cannot be understood as a limitation to the disclosure.

[0023] It is to be noted that, in the description of the disclosure, unless the context clearly indicates otherwise, the terms “mounted”, “coupled”, and “connected” should be broadly understood. For example, these terms may refer to a fixed connection, a removable connection, or an integrated connection; or, may refer to a mechanical connection; or, may refer to a direct connection, an indirect connection through an intermediary, or an internal communication or interaction of two elements. For those skilled in the art, the meanings of the above terms referred to in the disclosure may be understood based on specific situations. It is to be noted that, the terms “first”, “second”, and the like used in the specification, the claims, and the accompany drawings of the disclosure are used to distinguish different objects rather than describe a particular order.

[0024] In addition, the terms “include”, “may include”, “have”, or “may have” used in the disclosure indicate existence of disclosed corresponding functions, operations, elements, etc., and

do not constitute limitations on one or more additional functions, operations, elements, etc. Furthermore, the terms “include”, “comprise”, and “have” as well as variations thereof refer to presence of corresponding features, numbers, operations, elements, components, or combinations thereof disclosed in the specification, which does not exclude presence or addition of one or more other features, numbers, operations, elements, components, or combinations thereof, and are intended to cover non-exclusive inclusion. In addition, when describing the embodiments of the disclosure, the term “may” represents “one or more embodiments of the disclosure”. Moreover, the term “exemplary” is intended to refer to examples or illustrations.

[0025] Unless otherwise defined, all technical and scientific terms herein have the same meaning as commonly understood by one of ordinary skill in the technical field to which the disclosure belongs. The terms herein are merely for the purpose of describing embodiments of the disclosure, which are not intended to limit the disclosure.

[0026] In view of the above deficiencies of the related art, the disclosure provides a mass transfer assembly, a display panel, and a display device having the display panel, which aims to transfer micro LED elements on a growth substrate to a driving substrate at one time.

[0027] In order to solve the above technical problems, the disclosure provides a mass transfer assembly. The mass transfer assembly is used for transferring a plurality of light-emitting elements from a growth substrate to a driving substrate. Each of the plurality of light-emitting elements includes a light-emitting body connected with the growth substrate and two electrodes connected with the light-emitting body. The mass transfer assembly includes a temporary substrate and a transfer substrate. The temporary substrate is configured to adhere two electrodes of each of a plurality of light-emitting elements on the growth substrate, to separate the plurality of light-emitting elements from the growth substrate, such that a light-emitting body of each of the plurality of light-emitting elements is suspended. The transfer substrate has a plurality of adhesion regions, and the transfer substrate is configured to adhere, through the adhesion regions, a light-emitting body of each of a plurality of light-emitting elements on the temporary substrate, where one adhesion region adheres a light-emitting body of one light-emitting element, to separate the plurality of light-emitting elements from the temporary substrate, such that two electrodes of each of the plurality of light-emitting elements are suspended. The driving substrate has a plurality of binding regions, positions of the plurality of binding regions on the driving substrate are in one-to-one correspondence with positions of the plurality of adhesion regions on the transfer substrate. The transfer substrate is further configured to transfer the plurality of light-emitting elements to one side of the driving substrate, to bind the two electrodes of each of the plurality of light-emitting elements to a corresponding binding region of the driving substrate.

[0028] In sum, the mass transfer assembly provided in embodiments of the disclosure includes the temporary substrate and the transfer substrate, the positions of the plurality of binding regions on the driving substrate are in one-to-one correspondence with the positions of the plurality of adhesion regions on the transfer substrate, and the plurality of binding regions of the driving substrate are used for binding the light-emitting elements. As such, the plurality of light-emitting elements adhered to the transfer substrate can be transferred to the driving substrate at one time, thereby reducing workload and cost of transferring of the light-emitting elements.

[0029] In exemplary embodiments, the temporary substrate includes a first base substrate and an adhesion layer. The adhesion layer is disposed on a surface of the first base substrate. The adhesion layer is used for adhering the two electrodes of each of the plurality of light-emitting elements on the growth substrate.

[0030] In exemplary embodiments, the transfer substrate includes a second substrate, a plurality of photothermal elements, a packaging layer, a plurality of expansion media, a flexible layer, and a plurality of adhesion elements. The plurality of photothermal elements are spaced apart from one another on a surface of the second substrate. The packaging layer covers the plurality of photothermal elements on the second substrate, and the packaging layer defines a plurality of

accommodating holes penetrating through the packaging layer. Each of the plurality of accommodating holes is filled with an expansion medium, and the expansion medium is in contact with the photothermal element. The flexible layer shields and covers the plurality of expansion media. The plurality of adhesion elements are disposed on a surface of the flexible layer facing away from the plurality of expansion media. A position of one expansion medium and a position of one adhesion element both correspond to a position of one adhesion region. The photothermal element is configured to receive lights to generate heat. The expansion medium is configured to receive heat conducted by the photothermal element to expand to abut against the flexible layer, and the flexible layer abuts against the adhesion element, such that the adhesion element adheres to the light-emitting body of the light-emitting element.

[0031] In exemplary embodiments, the transfer substrate includes a flat layer, a plurality of electric heating elements, a packaging layer, a plurality of expansion media, a plurality of flexible layers, and a plurality of adhesion elements. The plurality of electric heating elements are disposed on a surface of the flat layer. The packaging layer covers the plurality of electric heating elements on the flat layer, and the packaging layer defines a plurality of accommodating holes penetrating through the packaging layer. Each of the plurality of accommodating holes is filled with the expansion medium, and the expansion medium is in contact with the electric heating element. The flexible layer covers the plurality of expansion media. The plurality of adhesion elements are disposed on a surface of the flexible layer facing away from the plurality of expansion media. A position of one expansion medium and a position of one adhesion element both correspond to a position of one adhesion region. The electric heating element is configured to receive a current to generate heat. The expansion medium is configured to receive heat conducted by the electric heating element to expand to abut against the flexible layer, and the flexible layer abuts against the adhesion element, such that the adhesion element adheres to the light-emitting body of the light-emitting element.

[0032] In exemplary embodiments, the transfer substrate further includes a plurality of control transistors. Each of the plurality of control transistors has a gate, an active device, a drain, and a source. The active device is electrically connected with the drain and the source. The drain of the control transistor is further electrically connected with one electric heating element. The source of the control transistor is further electrically connected with a power-supply terminal. The gate of the control transistor is configured to receive a control signal to turn on the active device of the control transistor, to make the drain of the control transistor be electrically connected with the source of the control transistor, such that the electric heating element receives a current outputted by the power-supply terminal.

[0033] In exemplary embodiments, the transfer substrate further includes a second substrate, an insulating layer, and a passivation layer. The second substrate is disposed on one side of the flat layer facing away from the plurality of electric heating elements, and is spaced apart from the flat layer. A plurality of gates are spaced apart from one another on one side of the second substrate facing the flat layer. The insulating layer covers the plurality of gates on the second substrate. A plurality of active devices are disposed on a surface of the insulating layer facing away from the second substrate. A position of one active device corresponds to a position of one gate. The drain and the source are connected with the active device. The drain is spaced apart from the source. The passivation layer covers the plurality of active devices, the plurality of drains, and the plurality of sources on the insulating layer. The passivation layer is connected with the flat layer.

[0034] In exemplary embodiments, the flat layer defines a plurality of first vias penetrating through the flat layer, the passivation layer defines a plurality of second vias penetrating through the passivation layer, one drain is exposed from one second via, and one second via communicates with one first via. The transfer substrate further includes a plurality of connecting electrodes, one connecting electrode is accommodated in one first via and one second via which are in communication, and one connecting electrode is connected with one electric heating element and one drain.

[0035] In exemplary embodiments, the expansion medium is made of nitrogen, argon, water, or alcohol, and the flexible layer is made of polyester fiber or polyvinyl alcohol.

[0036] Based on the same inventive concept, the disclosure further provides a display panel. The display panel includes a driving substrate and a plurality of light-emitting elements disposed on one side of the driving substrate. The plurality of light-emitting elements are transferred to the driving substrate by the above mass transfer assembly.

[0037] In sum, the display panel provided in embodiments of the disclosure includes the driving substrate and the plurality of light-emitting elements, the plurality of light-emitting elements are transferred to the driving substrate by the mass transfer assembly, the mass transfer assembly includes the temporary substrate and the transfer substrate, the positions of the plurality of binding regions on the driving substrate are in one-to-one correspondence with the positions of the plurality of adhesion regions on the transfer substrate, and the plurality of binding regions of the driving substrate are used for binding the light-emitting elements. As such, the plurality of light-emitting elements adhered to the transfer substrate can be transferred to the driving substrate at one time, thereby reducing workload and cost of transferring of the light-emitting elements.

[0038] Based on the same inventive concept, the disclosure further provides a display device. The display device includes a housing and the above display panel. The display panel is received in the housing. A light-emitting side of the display panel is exposed from the housing.

[0039] In sum, the display device provided in embodiments of the disclosure includes the housing and the display panel, the display panel includes the driving substrate and the plurality of light-emitting elements, the plurality of light-emitting elements are transferred to the driving substrate by the mass transfer assembly, the mass transfer assembly includes the temporary substrate and the transfer substrate, the positions of the plurality of binding regions on the driving substrate are in one-to-one correspondence with the positions of the plurality of adhesion regions on the transfer substrate, and the plurality of binding regions of the driving substrate are used for binding the light-emitting elements. As such, the plurality of light-emitting elements adhered to the transfer substrate can be transferred to the driving substrate at one time, thereby reducing workload and cost of transferring of the light-emitting elements.

[0040] Referring to FIG. 1, FIG. 1 is a schematic diagram illustrating a layer structure of a growth substrate and light-emitting elements provided in a first embodiment of the disclosure. As illustrated in FIG. 1, multiple light-emitting elements **100** are formed on a surface of a growth substrate **200**, and each light-emitting element **100** includes a light-emitting body **110** as well as a first electrode **120** and a second electrode **130** which are electrically connected with the light-emitting body **110**. The light-emitting body **110** is formed on the surface of the growth substrate **200**, and the first electrode **120** and the second electrode **130** each are formed on a surface of the light-emitting body **110** facing away from the growth substrate **200**, such that the first electrode **120** is suspended and the second electrode **130** is suspended. The expression “the first electrode **120** is suspended” means that one end of the first electrode **120** is connected with the light-emitting body **110** while the other end of the first electrode **120** facing away from the light-emitting body **110** is not connected with other components. The expression “the second electrode **130** is suspended” means that one end of the second electrode **130** is connected with the light-emitting body **110** while the other end of the second electrode **130** facing away from the light-emitting body **110** is not connected with other components.

[0041] In exemplary embodiments of the disclosure, the light-emitting element **100** may be a micro light-emitting diode (LED).

[0042] Referring to FIG. 2 and FIG. 3, FIG. 2 is a schematic structural diagram illustrating a temporary substrate of a mass transfer assembly provided in the first embodiment of the disclosure, and FIG. 3 is a schematic diagram illustrating adhering of light-emitting elements with the temporary substrate illustrated in FIG. 2. The mass transfer assembly provided in embodiments of the disclosure is used for transferring multiple light-emitting elements **100** on the growth substrate

200 to a driving substrate. The mass transfer assembly includes a temporary substrate **300**. The temporary substrate **300** is aligned with the growth substrate **200**, such that the first electrode **120** and the second electrode **130** of each light-emitting element **100** on the growth substrate **200** face the temporary substrate **300**. The temporary substrate **300** is configured to adhere the first electrode **120** and the second electrode **130** of each light-emitting element **100** on the growth substrate **200**, to separate the multiple light-emitting elements **100** from the growth substrate **200**, such that the light-emitting body **110** of each light-emitting element **100** is suspended. The light-emitting body **110** being suspended means that one side of the light-emitting body **110** is connected with the first electrode **120** and the second electrode **130** while the other side of the light-emitting body **110** facing away from the first electrode **120** and the second electrode **130** is not connected with other components.

[0043] Referring to FIG. 4 and FIG. 5, FIG. 4 is a schematic diagram illustrating a first structure of a transfer substrate of the mass transfer assembly provided in the first embodiment of the disclosure, and FIG. 5 is a schematic diagram illustrating adhering of light-emitting elements with the transfer substrate illustrated in FIG. 4. The mass transfer assembly further includes a transfer substrate **400**. The transfer substrate **400** has multiple adhesion regions **400a** arranged in an array and spaced apart from one another. The transfer substrate **400** is aligned with the temporary substrate **300**, such that the light-emitting body **110** of each light-emitting element **100** on the temporary substrate **300** faces the transfer substrate **400**. The transfer substrate **400** is configured to adhere the light-emitting body **110** of each light-emitting element **100** on the temporary substrate **300**, where one adhesion region **400a** adheres the light-emitting body **110** of one light-emitting element **100**, to separate the multiple light-emitting elements **100** from the temporary substrate **300**, such that the first electrode **120** and the second electrode **130** of each light-emitting element **100** are suspended.

[0044] Referring to FIG. 6, FIG. 6 is a schematic diagram illustrating transferring of light-emitting elements to the driving substrate from the transfer substrate illustrated in FIG. 4. A driving substrate **600** illustrated in FIG. 6 has multiple binding regions **600a** arranged in an array and spaced apart from one another. Positions of the multiple binding regions **600a** on the driving substrate **600** are in one-to-one correspondence with positions of the multiple adhesion regions **400a** on the transfer substrate **400**, that is, the positions of the multiple binding regions **600a** of the driving substrate **600** are in one-to-one correspondence with the positions of the multiple adhesion regions **400a** of the transfer substrate **400**. The transfer substrate **400** is aligned with the driving substrate **600**, such that the multiple binding regions **600a** coincide with the multiple adhesion regions **400a** in one-to-one correspondence, and the first electrode **120** and the second electrode **130** of each light-emitting element **100** on the transfer substrate **400** face the driving substrate **600**. The transfer substrate **400** is further configured to transfer the multiple light-emitting elements **100** to one side of the driving substrate **600**. The driving substrate **600** fixes the first electrode **120** and the second electrode **130** of each light-emitting element **100**, such that the first electrode **120** and the second electrode **130** of each light-emitting element **100** are fixed to a corresponding binding region **600a** of the driving substrate **600**.

[0045] It can be understood that, the multiple binding regions **600a** of the driving substrate **600** are used for binding the light-emitting elements **100**, and a distance between two adjacent light-emitting elements **100** on the growth substrate **200** is smaller than a distance between two adjacent binding regions **600a** of the driving substrate **600**. The positions of the multiple binding regions **600a** on the driving substrate **600** are in one-to-one correspondence with the positions of the multiple adhesion regions **400a** on the transfer substrate **400**, that is, a distance between two adjacent adhesion regions **400a** is consistent with the distance between two adjacent binding regions **600a**. As such, multiple light-emitting elements **100** adhered to the transfer substrate **400** can be transferred to the driving substrate **600** at one time, thereby reducing workload and cost of transferring of the light-emitting elements **100**. Before the first electrode **120** and the second

electrode **130** of the light-emitting element **100** are bound to the driving substrate **600**, the first electrode **120** and the second electrode **130** of each light-emitting element **100** are suspended by sequentially using the temporary substrate **300** and the transfer substrate **400**, and then the first electrode **120** and the second electrode **130** of each light-emitting element **100** are in contact with the driving substrate **600** for binding.

[0046] In embodiments of the disclosure, referring to FIG. 2 and FIG. 3, the temporary substrate **300** includes a first base substrate **310** and an adhesion layer **320**, and the adhesion layer **320** is disposed on a surface of the first base substrate **310**. The temporary substrate **300** is aligned with the growth substrate **200**, the adhesion layer **320** of the temporary substrate **300** faces the growth substrate **200**, the first electrode **120** and the second electrode **130** of each light-emitting element **100** on the growth substrate **200** face the adhesion layer **320** of the temporary substrate **300**, and the adhesion layer **320** is connected with the first electrode **120** and the second electrode **130** of the multiple light-emitting elements **100**. The adhesion layer **320** is used for adhering the first electrode **120** and the second electrode **130** of each light-emitting element **100**.

[0047] It can be understood that, the light-emitting body **110** of each light-emitting element **100** may be separated from the growth substrate **200** through a laser lift off (LLO) process, such that the light-emitting body **110** of each light-emitting element **100** is suspended. In exemplary embodiments of the disclosure, the adhesion layer **320** is an adhesive glue.

[0048] In embodiments of the disclosure, referring to FIG. 4, the transfer substrate **400** includes a flat layer **410**, multiple electric heating elements **420**, a packaging layer **430**, and multiple expansion media **440**. The multiple electric heating elements **420** are arranged in an array and spaced apart from one another on a surface of the flat layer **410**. The position of one electric heating element **420** corresponds to the position of one adhesion region **400a**, that is, an orthographic projection of one electric heating element **420** on the flat layer **410** falls into an orthographic projection of one adhesion region **400a** on the flat layer **410**. The packaging layer **430** is disposed on a surface of the multiple electric heating elements **420** facing away from the flat layer **410** and covers the flat layer **410**, that is, the packaging layer **430** covers the multiple electric heating elements **420** on the flat layer **410**. The packaging layer **430** defines multiple accommodating holes **430a** penetrating through the packaging layer **430**. The position of one accommodating hole **430a** corresponds to the position of one electric heating element **420**, that is, an orthographic projection of one accommodating hole **430a** on the flat layer **410** overlaps an orthographic projection of one electric heating element **420** on the flat layer **410**, such that a portion of the electric heating element **420** is exposed to the outside of an opening of the accommodating hole **430a** facing the electric heating element **420**. Each of the multiple accommodating holes **430a** is filled with an expansion medium **440**, and the expansion medium **440** is in contact with the electric heating element **420**. The flat layer **410** provides a flat surface, such that the multiple electric heating elements **420** are located on the same plane. The packaging layer **430** provides the multiple accommodating holes **430a** for accommodating the expansion media **440**.

[0049] In embodiments of the disclosure, referring to FIG. 4, the transfer substrate **400** further includes multiple flexible layers **450** and multiple adhesion elements **460**. The flexible layer **450** is disposed on a surface of the packaging layer **430** facing away from the flat layer **410** and covers the multiple expansion media **440**. The multiple adhesion elements **460** are disposed on a surface of the flexible layer **450** facing away from the multiple expansion media **440**. The position of one adhesion element **460** corresponds to the position of one expansion medium **440**, that is, an orthographic projection of one adhesion element **460** on the flat layer **410** overlaps an orthographic projection of one expansion medium **440** on the flat layer **410**, and the orthographic projection of one adhesion element **460** on the flat layer **410** overlaps an orthographic projection of one adhesion region **400a** on the flat layer **410**. Referring to FIG. 5, the transfer substrate **400** is aligned with the temporary substrate **300**, the multiple adhesion elements **460** of the transfer substrate **400** face the

temporary substrate **300**, the light-emitting body **110** of each of multiple light-emitting elements **100** on the temporary substrate **300** faces the transfer substrate **400**, and the adhesion elements **460** is spaced apart from the light-emitting elements **100** by a preset distance. The electric heating element **420** is configured to generate heat after receiving a current. Since the expansion medium **440** is in contact with the electric heating element **420**, the expansion medium **440** can expand to abut against the flexible layer **450** after receiving the heat conducted by the electric heating element **420**, such that a portion of the flexible layer **450** connected with the expansion medium **440** protrudes toward one side of the flexible layer **450** facing away from the packaging layer **430** (that is, protruding toward the adhesion element **460**) to abut against the adhesion element **460**. A portion of the expansion medium **440** protrudes toward one side of the flexible layer **450** facing away from the packaging layer **430**, such that the height of the adhesion element **460** increases. The adhesion element **460** with the increased height is configured to adhere the light-emitting body **110** of the light-emitting element **100**. The connection strength between the adhesion element **460** and the light-emitting element **100** is greater than the connection strength between the adhesion layer **320** and the light-emitting element **100**. A surface of the flat layer **410** facing the packaging layer **430** is a reference plane for the height, and the height increases in a direction in which the packaging layer **430** faces away from the flat layer **410**.

[0050] It can be understood that, after the electric heating element **420** receives the current, the adhesion element **460** corresponding to the position of the electric heating element **420** can adhere the light-emitting element **100**, and the position of the adhesion element **460** corresponds to the position of the adhesion region **400a**, as such, the light-emitting element(s) **100** adhered to the transfer substrate **400** is only located in the adhesion region **400a**. The multiple electric heating elements **420** can selectively receive the current, such that some of the multiple electric heating elements **420** receive the current while the remaining electric heating elements **420** do not receive the current, and further, some of the multiple adhesion elements **460** adhere the light-emitting body **110** of the light-emitting element **100** while the remaining adhesion elements **460** do not adhere the light-emitting element **100**, that is, the multiple adhesion elements **460** can selectively adhere the light-emitting elements **100**, as such, the adhesion element(s) **460** located at a specified position(s) of the transfer substrate **400** adheres the light-emitting element(s) **100** on the temporary substrate **300**. Since the connection strength between the adhesion element **460** and the light-emitting element **100** is greater than the connection strength between the adhesion layer **320** and the light-emitting element **100**, the light-emitting element(s) **100** adhered to the transfer substrate **400** can be separated from the temporary substrate **300** under action of an external force.

[0051] It is to be noted that, the driving substrate **600** and multiple light-emitting elements **100** on the driving substrate **600** are part of a display panel. The multiple light-emitting elements **100** on the driving substrate **600** have three types, that is, multiple first light-emitting elements for emitting red lights, multiple second light-emitting elements for emitting green lights, and multiple third light-emitting elements for emitting blue lights. The three types of light-emitting elements (i.e., the multiple first light-emitting elements, the multiple second light-emitting elements, and the multiple third light-emitting elements) may be respectively formed on different growth substrates **200**. By selectively adhering light-emitting elements **100** with the multiple adhesion elements **460**, the transfer substrate **400** can adhere the multiple first light-emitting elements, the multiple second light-emitting elements, and the multiple third light-emitting elements at the same time, and then the transfer substrate **400** can transfer the multiple first light-emitting elements, the multiple second light-emitting elements, and the multiple third light-emitting elements to the driving substrate **600** at one time, thereby improving efficiency of mass transfer. Furthermore, if a certain light-emitting element **100** on the growth substrate **200** is damaged, the damaged light-emitting element **100** will be adhered to the temporary substrate **300**, but an adhesion element **460** corresponding to the position of the damaged light-emitting element **100** will not adhere the damaged light-emitting element **100**, thereby avoiding the damaged light-emitting element **100** from being transferred to

the driving substrate **600**.

[0052] In embodiments of the disclosure, referring to FIG. 6, the driving substrate **600** includes a driving circuit layer **610**, multiple first conductive elements **620**, and multiple second conductive elements **630**. The multiple first conductive elements **620** and the multiple second conductive elements **630** are disposed on one side of the driving circuit layer **610**. One first conductive element **620** and one second conductive element **630** are disposed in one binding region **600a**. The transfer substrate **400** is aligned with the driving substrate **600**, such that multiple binding regions **600a** coincide with multiple adhesion regions **400a** in one-to-one correspondence, the first electrode **120** and the second electrode **130** of each light-emitting element **100** on the transfer substrate **400** face the driving substrate **600**, and the multiple first conductive elements **620** and the multiple second conductive elements **630** of the driving substrate **600** face the transfer substrate **400**. The first electrode **120** of the light-emitting element **100** is in contact with the first conductive element **620**, and the second electrode **130** of the light-emitting element **100** is in contact with the second conductive element **630**, such that the first electrode **120** is bound to the first conductive element **620** and the second electrode **130** is bound to the second conductive element **630**. The connection strength between the driving substrate **600** and the light-emitting element **100** is greater than the connection strength between the transfer substrate **400** and the light-emitting element **100**, such that the light-emitting element **100** fixed to the driving substrate **600** can be separated from the transfer substrate **400** under action of an external force.

[0053] It can be understood that, as illustrated in FIG. 6, an electric heating element **420** is configured to receive a current, and accordingly, the height of a light-emitting element **100** corresponding to the position of the electric heating element **420** increases, such that the light-emitting element **100** corresponding to the position of the electric heating element **420** moves toward the driving substrate **600** and is bound to the driving substrate **600**. The multiple electric heating elements **420** can selectively receive the current, such that the height of some light-emitting elements **100** adhered to the transfer substrate **400** increases, and the light-emitting elements **100** with the increased height is in contact with and bound to the driving substrate **600**, while the light-emitting element(s) **100** of which the height is not increased is not in contact with the driving substrate **600** and is therefore not bound. As such, by making the multiple electric heating elements **420** receive the current selectively, the light-emitting element **100** can be bound to the binding region **600a** at a specified position of the driving substrate **600**.

[0054] It is to be noted that, the light-emitting elements **100** adhered to the transfer substrate **400** may also be of the same type, that is, the light-emitting elements **100** adhered to the transfer substrate **400** may all be the first light-emitting elements, or may all be the second light-emitting elements, or may all be the third light-emitting elements, and the multiple light-emitting elements **100** on the driving substrate **600** include multiple first light-emitting elements, multiple second light-emitting elements, and multiple third light-emitting elements. The multiple electric heating elements **420** can selectively receive the current, such that the height of the light-emitting element **100** at a specified position among multiple light-emitting elements **100** of the same type on the transfer substrate **400** increases, to make the light-emitting elements **100** of the same type be bound to the binding regions **600a** at the specified positions of the driving substrate **600**.

[0055] In exemplary embodiments, after the light-emitting elements **100** are transferred to and bound to the driving substrate **600** from the transfer substrate **400**, the multiple electric heating elements **420** stop receiving the current, and accordingly, the temperature of the expansion medium **440** drops to the room temperature, and the volume of the expansion medium **440** returns to its original volume for a next transfer.

[0056] In exemplary embodiments, the material of the flat layer **410** may include a polyimide (PI) film or a polymethyl methacrylate (PMMA), which is not limited in the disclosure. The material of the expansion medium **440** may include nitrogen, argon, water, alcohol, or other inert gases that expand when heated, which is not limited in the disclosure. The electric heating element **420**, the

packaging layer **430**, and the flexible layer **450** cooperate to seal the expansion medium **440** filled in the accommodating hole **430a**. The material of the flexible layer **450** may include a polyester film (PET), a polyvinyl alcohol (PVA) film, or other organic film layers with a good deformation property, which is not limited in the disclosure. The adhesion element **460** may be an adhesive glue.

[0057] In embodiments of the disclosure, referring to FIG. 4 and FIG. 5, the transfer substrate **400** further includes a second substrate **480**, multiple control transistors **490**, an insulating layer **510**, and a passivation layer **520**. The second substrate **480** is disposed on one side of the flat layer **410** facing away from the multiple electric heating elements **420** and is spaced apart from the flat layer **410**. Each control transistor **490** includes a gate **491**, an active device **492**, a drain **493**, and a source **494**. Multiple gates **491** are spaced apart from one another on one side of the second substrate **480** facing the flat layer **410**. An orthographic projection of one electric heating element **420** on the second substrate **480** is adjacent to an orthographic projection of one gate **491** on the second substrate **480**. The insulating layer **510** is disposed on the surface of each gate **491** facing away from the second substrate **480** and covers the second substrate **480**, that is, the insulating layer **510** covers the multiple gates **491** on the second substrate **480**. Multiple active devices **492** are disposed on a surface of the insulating layer **510** facing away from the second substrate **480**. The position of one active device **492** corresponds to the position of one gate **491**, that is, an orthographic projection of one active device **492** on the second substrate **480** overlaps or partially overlaps an orthographic projection of one gate **491** on the second substrate **480**. The drain **493** and the source **494** are disposed on a portion of circumference of the active device **492** and a portion of a surface of the active device **492** facing away from the insulating layer **510**, and the drain **493** is spaced apart from the source **494**. The active device **492** is electrically connected with the drain **493** and the source **494**. The passivation layer **520** is disposed on a surface of the insulating layer **510** facing away from the second substrate **480**, and covers the multiple active devices **492**, the multiple drains **493**, and the multiple sources **494**, that is, the passivation layer **520** covers the multiple active devices **492**, the multiple drains **493**, and the multiple sources **494** on the insulating layer **510**. A surface of the passivation layer **520** facing away from the insulating layer **510** is connected with a surface of the flat layer **410** facing away from the multiple electric heating elements **420**. The insulating layer **510** is used for insulating the gate **491** from the active device **492**. The passivation layer **520** is used for isolating the active device **492**, the drain **493**, and the source **494**, to prevent impurities such as water and oxygen from entering the active device **492**, the drain **493**, and the source **494**. The drain **493** is electrically connected with the electric heating element **420**, and the source **494** is electrically connected with a power-supply terminal for providing a current to the electric heating element **420**. The gate **491** is configured to receive a control signal to turn on the active device **492**, to make the drain **493** and the source **494** conductive through the active device **492**, such that the power-supply terminal can output a current to the electric heating element **420**.

[0058] It can be understood that, if the gate **491** of the control transistor **490** receives the control signal, the electric heating element **420** electrically connected with the drain **493** of the control transistor **490** receives the current. If no control signal is received by the gate **491** of the control transistor **490**, no current is received by the electric heating element **420** electrically connected with the drain **493** of the control transistor **490**. As such, the multiple electric heating elements **420** can selectively receive the current.

[0059] In exemplary embodiments, the control transistor **490** may be a thin-film transistor (TFT) or a metal oxide semiconductor field effect transistor (MOSFET). The material of the gate **491**, the material of the drain **493**, and the material of the source **494** each may include metal materials such as aluminum, molybdenum, or copper. The material of the active device **492** may include single crystal silicon, polycrystalline silicon, or indium gallium zinc oxide, etc. The material of the insulating layer **510** and the material of the passivation layer **520** each may include silicon nitride

or silicon oxide, etc. The disclosure does not limit the material of the above structure.

[0060] In exemplary embodiments, referring to FIG. 4, the flat layer **410** defines multiple first vias **410a** penetrating through the flat layer **410**. The position of one first via **410a** corresponds to the position of one drain **493**, that is, an orthographic projection of one first via **410a** on the second substrate **480** overlaps or partially overlaps an orthographic projection of one drain **493** on the second substrate **480**. The passivation layer **520** defines multiple second vias **520a** penetrating through the passivation layer **520**. The position of one second via **520a** corresponds to the position of one drain **493**, that is, an orthographic projection of one second via **520a** on the second substrate **480** overlaps or partially overlaps an orthographic projection of one drain **493** on the second substrate **480**. A portion of the drain **493** is exposed from one second via **520a**, and one second via **520a** communicates with one first via **410a**, where the expression “communicate with” means connecting and communicating. The transfer substrate **400** further includes multiple connecting electrodes **530**. A portion of one connecting electrode **530** is accommodated in a first via **410a** and a second via **520a** and is connected with the drain **493**, and another portion of the connecting electrode **530** is disposed on a portion of a surface of the flat layer **410** facing away from the passivation layer **520** and is connected with an electric heating element **420**, such that the electric heating element **420** is electrically connected with the drain **493**. That is, one end of the connecting electrode **530** is connected with the electric heating element **420**, and another end of the connecting electrode **530** extends through the first via **410a** and the second via **520a** to be connected to the drain **493**, such that the electric heating element **420** is electrically connected with the drain **493** through the connecting electrode **530**. A portion of the packaging layer **430** is filled in the first vias **410a** and the second vias **520a**.

[0061] In exemplary embodiments, the material of the connecting electrode **530** includes indium tin oxide (ITO).

[0062] Referring to FIG. 7, FIG. 7 is a schematic diagram illustrating a second structure of the transfer substrate of the mass transfer assembly provided in the first embodiment of the disclosure. The difference between the transfer substrate of the second structure and the transfer substrate of the first structure is that: the transfer substrate of the second structure does not include the flat layer **410**, the multiple electric heating elements **420**, the multiple control transistors **490**, the insulating layer **510**, and the multiple passivation layers **520**. The transfer substrate of the second structure includes the second substrate **480**, multiple photothermal elements **540**, the packaging layer **430**, the multiple expansion media **440**, the flexible layer **450**, and the multiple adhesion elements **460**. For details of the structures of the transfer substrate of the second structure and the transfer substrate of the first structure, reference can be made to the relevant description of the transfer substrate of the first structure, which will not be repeated herein.

[0063] Specifically, referring to FIG. 7, the multiple photothermal elements **540** are disposed on a surface of the second substrate **480**, and the multiple photothermal elements **540** are spaced apart from one another. The packaging layer **430** is disposed on the surface of each of the multiple photothermal elements **540** facing away from the second substrate **480**, and covers a surface of the second substrate **480** on which the multiple photothermal elements **540** are disposed, that is, the packaging layer **430** covers the multiple photothermal elements **540** on the second substrate **480**. The packaging layer **430** defines multiple accommodating holes **430a** penetrating through the packaging layer **430**. Each of the multiple accommodating holes **430a** is filled with the expansion medium **440**. The flexible layer **450** is disposed on a surface of the packaging layer **430** facing away from the second substrate **480**, and shields and covers the multiple expansion media **440**. The multiple adhesion elements **460** are disposed on a surface of the flexible layer **450** facing away from the multiple expansion media **440**. The position of one adhesion element **460** corresponds to the position of one expansion medium **440**, that is, an orthographic projection of one adhesion element **460** on the second substrate **480** overlaps or partially overlaps an orthographic projection of one expansion medium **440** on the second substrate **480**.

[0064] In embodiments of the disclosure, referring to FIG. 8, FIG. 8 is a schematic diagram illustrating adhering of light-emitting elements with the transfer substrate illustrated in FIG. 7. The photothermal element **540** is configured to generate heat after receiving lights. Since the expansion medium **440** is in contact with the photothermal element **540**, the expansion medium **440** can expand to abut against the flexible layer **450** after receiving the heat conducted by the photothermal element **540**, such that a portion of the flexible layer **450** connected with the expansion medium **440** protrudes toward one side of the flexible layer **450** facing away from the packaging layer **430** (i.e., protruding toward the adhesion element **460**) to abut against the adhesion element **460**. A portion of the expansion medium **440** protrudes toward the side of the flexible layer **450** facing away from the packaging layer **430**, such that the height of the adhesion element **460** increases. The adhesion element **460** with the increased height is configured to adhere the light-emitting body **110** of the light-emitting element **100**. Regarding the height, a surface of the second substrate **480** facing the packaging layer **430** is taken as a reference plane, and a direction in which the packaging layer **430** faces away from the second substrate **480** is a direction in which the height increases. The multiple photothermal elements **540** can selectively receive lights, such that the multiple adhesion elements **460** can selectively adhere the light-emitting body **110** of the light-emitting element **100**.

[0065] In embodiments of the disclosure, referring to FIG. 9, FIG. 9 is a schematic diagram illustrating transferring of light-emitting elements to the driving substrate from the transfer substrate illustrated in FIG. 7. The photothermal element **540** is configured to receive lights, and the height of a light-emitting element **100** corresponding to the position of the photothermal element **540** increases, such that the light-emitting element **100** corresponding to the position of the photothermal element **540** moves toward the driving substrate **600** and is bound to the driving substrate **600**. The multiple photothermal elements **540** can selectively receive lights, such that the height of some light-emitting elements **100** adhered to the transfer substrate **400** increases, and the light-emitting elements **100** of which the height is increased is in contact with and bound to the driving substrate **600**, while the light-emitting element(s) **100** of which the height is not increased is not in contact with the driving substrate **600** and is therefore not bound. As such, by making the multiple photothermal elements **540** selectively receive lights, the light-emitting element **100** can be bound to the binding region **600a** at a specified position of the driving substrate **600**.

[0066] In exemplary embodiments, the material of the photothermal element **540** includes PI containing carbon powder, PMMA containing carbon powder, or metal chromium, which is not limited in the disclosure.

[0067] In sum, the mass transfer assembly provided in embodiments of the disclosure includes the temporary substrate **300** and the transfer substrate **400**. The temporary substrate **300** is configured to adhere the first electrode **120** and the second electrode **130** of each light-emitting element **100** on the growth substrate **200**, to separate the multiple light-emitting elements **100** from the growth substrate **200**, such that the light-emitting body **110** of each light-emitting element **100** is suspended. The transfer substrate **400** has multiple adhesion regions **400a** arranged in an array and spaced apart from one another. The transfer substrate **400** is configured to adhere the light-emitting body **110** of each light-emitting element **100** on the temporary substrate **300**, where the light-emitting body **110** of one light-emitting element **100** is adhered to one adhesion region **400a**, to separate the multiple light-emitting elements **100** from the temporary substrate **300**, such that the first electrode **120** and the second electrode **130** of each of the multiple light-emitting elements **100** are suspended. The driving substrate **600** has multiple binding regions **600a** arranged in an array and spaced apart from one another, and positions of the multiple binding regions **600a** are in one-to-one correspondence with positions of the multiple adhesion regions **400a**. The transfer substrate **400** is further configured to transfer the multiple light-emitting elements **100** to one side of the driving substrate **600**. The driving substrate **600** fixes the first electrode **120** and the second electrode **130** of each of the multiple light-emitting elements **100**, such that the first electrode **120**

and the second electrode **130** of each light-emitting element **100** are fixed to a corresponding binding region **600a** of the driving substrate **600**. As such, positions of the multiple binding regions **600a** of the driving substrate **600** are in one-to-one correspondence with positions of the multiple adhesion regions **400a** of the transfer substrate **400**, that is, a distance between two adjacent adhesion regions **400a** is identical with a distance between two adjacent binding regions **600a**, and thus, the multiple light-emitting elements **100** adhered to the transfer substrate **400** can be transferred to the driving substrate **600** at one time, thereby reducing workload and cost of transferring of the light-emitting elements **100**.

[0068] Based on the same inventive concept, the second embodiment of the disclosure provides a display panel. Referring to FIG. **10**, FIG. **10** is a schematic diagram illustrating a layer structure of a display panel provided in the second embodiment of the disclosure. A display panel **800** provided in embodiments of the disclosure includes a driving substrate **600** and multiple light-emitting elements **100**. The multiple light-emitting elements **100** are arranged on one side of the driving substrate **600** and are electrically connected with the driving substrate **600**. The multiple light-emitting elements **100** are transferred to the driving substrate **600** by the above mass transfer assembly. The driving substrate **600** is configured to transmit an electrical signal to the multiple light-emitting elements **100** to control the multiple light-emitting elements **100** to emit lights.

[0069] In exemplary embodiments, the first electrode **120** of each of the multiple light-emitting elements **100** is bound to one first conductive element **620** of the driving substrate **600**, such that the first electrode **120** is electrically connected with the first conductive element **620**. The second electrode **130** of each of the light-emitting elements **100** is bound to one second conductive elements **630** of the driving substrate **600**, such that the second electrode **130** is electrically connected with the second conductive element **630**.

[0070] In exemplary embodiments, the display panel **800** further includes multiple sub-pixel regions for emitting lights, and the position of one sub-pixel region corresponds to the position of one light-emitting element **100**, that is, the position of one sub-pixel region corresponds to the position of one binding region **600a** of the driving substrate **600**.

[0071] It can be understood that, the display panel **800** is applicable to an electronic device having functions such as personal digital assistants (PDA) and/or music players, for example, a mobile phone, a tablet computer, a wearable electronic device with a wireless communication function (e.g., a smart watch), etc. The above electronic device may also be other electronic devices, such as a laptop computer with a touch-sensitive surface (e.g., a touch panel), etc. In some embodiments, the electronic device may have a communication function, that is, the electronic device can establish communication with a network through the second-generation (2G) mobile phone communication technology specifications, the third-generation (3G) mobile phone communication technology specifications, the fourth-generation (4G) mobile phone communication technology specifications, the fifth-generation (5G) mobile phone communication technology specifications, the sixth-generation (6G) mobile phone communication technology specifications, a wireless local area network (W-LAN), or a communication method that may appear in the future. For the sake of simplicity, embodiments of the disclosure are not further limited.

[0072] Based on the same inventive concept, embodiments of the disclosure further provide a display device. Referring to FIG. **11**, FIG. **11** is a schematic diagram illustrating a layer structure of a display device provided in a third embodiment of the disclosure. A display device **1000** provided in embodiments of the disclosure includes a housing **900** and the above display panel **800**. The display panel **800** is received in the housing **900**. A light-emitting side of the display panel **800** is exposed from the housing **900**.

[0073] It can be understood that, the display device **1000** is applicable to an electronic device, which includes, but not limited to, a television, a tablet computer, a laptop computer, a desktop computer, a mobile phone, a vehicle-mounted display, a smart watch, a smart bracelet, smart glasses, etc. According to embodiments of the disclosure, the type of the display device **1000** is not

limited, and those skilled in the art may make corresponding designs according to a specific use requirement of the display device **1000**, which will not be repeated herein.

[0074] In exemplary embodiments, the display device **1000** may further include other necessary components and parts such as a power board, a high voltage board, and a key control board, and those skilled in the art may make corresponding supplements according to the specific type and actual functions of the display device **1000**, which will not be repeated herein.

[0075] In other embodiments of the disclosure, the display device **1000** may further include a processor and a memory. The processor is electrically connected with the display panel **800** and is configured to control display of the display panel **800**. The memory is electrically connected with the processor and is used to store program codes required for running of the processor, to control display contents of the display panel **800**, etc.

[0076] In exemplary embodiments, the memory may be a volatile memory, such as a random access memory (RAM). The memory may also be a non-volatile memory (NVM), such as a read-only memory (ROM), a flash memory (FM), a hard disk drive (HDD), or a solid-state drive (SSD). The memory may also be a combination of the above-mentioned types of memory.

[0077] In exemplary embodiments, the processor includes one or more general-purpose processors, where a general-purpose processor can be any type of device capable of processing electronic instructions, including a central processing unit (CPU), a microprocessor, a microcontroller, a main processor, a controller, etc. The processor is configured to execute various types of digital storage instructions, such as software or firmware programs stored in the memory, which enables a computing device to provide a wide variety of services.

[0078] It is to be understood that, the terms “first”, “second”, etc. are used for descriptive purposes only, and should not be understood as indicating or implying relative importance or implicitly indicating the number of technical features indicated. Thus, features defined as “first” or “second” may explicitly or implicitly include one or more of the features. In the description of embodiments of the disclosure, the meaning of “plurality/multiple” refers to two or more, unless otherwise clearly and specifically defined.

[0079] In the description of the specification, the description with reference to the terms “one embodiment”, “some embodiments”, “illustrative embodiments”, “examples”, “specific examples”, “some examples”, or the like means that specific features, structures, or characteristics described in conjunction with embodiments or examples are included in at least one embodiment or example of the disclosure. In the specification, the schematic representation of the above terms does not necessarily refer to the same embodiment or example. Moreover, the specific features, structures, or characteristics described may be combined in any one or more embodiments or examples in a suitable manner.

[0080] It should be understood that, the application of the disclosure is not limited to the foregoing exemplary embodiments. Those of ordinary skill in the art may make improvements or equivalent substitutions to the disclosure according to the above descriptions, and all these improvements and equivalent substitutions, however, shall all be encompassed within the protection scope of the appended claims of the disclosure. Those of ordinary skill in the art can understand that, all or part of the processes of the above embodiments can be implemented, and equivalent changes made according to the claims of the disclosure are still within the scope of the disclosure.

Claims

1. A mass transfer assembly for transferring a plurality of light-emitting elements from a growth substrate to a driving substrate, each of the plurality of light-emitting elements comprising a light-emitting body initially connected with the growth substrate and two electrodes connected with the light-emitting body, the mass transfer assembly comprising: a temporary substrate, wherein the temporary substrate is configured to adhere two electrodes of each of a plurality of light-emitting

elements on the growth substrate, to separate the plurality of light-emitting elements from the growth substrate, such that a light-emitting body of each of the plurality of light-emitting elements is suspended; and a transfer substrate having a plurality of adhesion regions, wherein the transfer substrate is configured to adhere, through the adhesion regions, a light-emitting body of each of a plurality of light-emitting elements on the temporary substrate, to separate the plurality of light-emitting elements from the temporary substrate, such that two electrodes of each of the plurality of light-emitting elements are suspended; wherein one adhesion region adheres a light-emitting body of one light-emitting element; wherein positions of the plurality of adhesion regions on the transfer substrate are in one-to-one correspondence with positions of a plurality of binding regions on the driving substrate; wherein the transfer substrate is further configured to transfer a plurality of light-emitting elements on the transfer substrate to one side of the driving substrate, to bind two electrodes of each of the plurality of light-emitting elements to a corresponding binding region of the driving substrate.

2. The mass transfer assembly of claim 1, wherein the temporary substrate comprises a first base substrate and an adhesion layer, the adhesion layer is disposed on a surface of the first base substrate, and the adhesion layer is used for adhering the two electrodes of each of the plurality of light-emitting elements on the growth substrate.

3. The mass transfer assembly of claim 1, wherein the transfer substrate comprises a second substrate, a plurality of photothermal elements, a packaging layer, a plurality of expansion media, a flexible layer, and a plurality of adhesion elements, wherein the plurality of photothermal elements are spaced apart from one another on a surface of the second substrate; the packaging layer covers the plurality of photothermal elements on the second substrate, and the packaging layer defines a plurality of accommodating holes penetrating through the packaging layer; each of the plurality of accommodating holes is filled with an expansion medium, and the expansion medium is in contact with the photothermal element; the flexible layer shields and covers the plurality of expansion media, the plurality of adhesion elements are disposed on a surface of the flexible layer facing away from the plurality of expansion media, and a position of one expansion medium and a position of one adhesion element both correspond to a position of one adhesion region; the photothermal element is configured to receive lights to generate heat; and the expansion medium is configured to receive heat conducted by the photothermal element to expand to abut against the flexible layer, and the flexible layer abuts against the adhesion element, such that the adhesion element adheres to the light-emitting body of the light-emitting element.

4. The mass transfer assembly of claim 1, wherein the transfer substrate comprises a flat layer, a plurality of electric heating elements, a packaging layer, a plurality of expansion media, a plurality of flexible layers, and a plurality of adhesion elements, wherein the plurality of electric heating elements are disposed on a surface of the flat layer; the packaging layer covers the plurality of electric heating elements on the flat layer, and the packaging layer defines a plurality of accommodating holes penetrating through the packaging layer; each of the plurality of accommodating holes is filled with the expansion medium, and the expansion medium is in contact with the electric heating element; the flexible layer covers the plurality of expansion media, the plurality of adhesion elements are disposed on a surface of the flexible layer facing away from the plurality of expansion media, and a position of one expansion medium and a position of one adhesion element both correspond to a position of one adhesion region; the electric heating element is configured to receive a current to generate heat; and the expansion medium is configured to receive heat conducted by the electric heating element to expand to abut against the flexible layer, and the flexible layer abuts against the adhesion element, such that the adhesion element adheres to the light-emitting body of the light-emitting element.

5. The mass transfer assembly of claim 4, wherein the transfer substrate further comprises a plurality of control transistors, wherein each of the plurality of control transistors has a gate, an active device, a drain, and a source, and the active device is electrically connected with the drain

and the source; the drain of the control transistor is further electrically connected with one electric heating element; the source of the control transistor is further electrically connected with a power-supply terminal; and the gate of the control transistor is configured to receive a control signal to turn on the active device of the control transistor, to make the drain of the control transistor be electrically connected with the source of the control transistor, such that the electric heating element receives a current outputted by the power-supply terminal.

6. The mass transfer assembly of claim 5, wherein the transfer substrate further comprises a second substrate, an insulating layer, and a passivation layer, wherein the second substrate is disposed on one side of the flat layer facing away from the plurality of electric heating elements and is spaced apart from the flat layer; gates of the plurality of control transistors are spaced apart from one another on one side of the second substrate facing the flat layer; the insulating layer covers the gates of the plurality of control transistors on the second substrate; active devices of the plurality of control transistors are disposed on a surface of the insulating layer facing away from the second substrate, a position of one active device corresponds to a position of one gate, the drain and the source are connected with the active device, and the drain is spaced apart from the source; and the passivation layer covers the active devices, drains, and sources of the plurality of control transistors on the insulating layer, and the passivation layer is connected with the flat layer.

7. The mass transfer assembly of claim 6, wherein the flat layer defines a plurality of first vias penetrating through the flat layer, the passivation layer defines a plurality of second vias penetrating through the passivation layer, one drain is exposed from one second via, and one second via communicates with one first via; and the transfer substrate further comprises a plurality of connecting electrodes, one connecting electrode is accommodated in one first via and one second via which are in communication, and one connecting electrode is connected with one electric heating element and one drain.

8. The mass transfer assembly of claim 3, wherein the expansion medium is made of nitrogen, argon, water, or alcohol, and the flexible layer is made of polyester fiber or polyvinyl alcohol.

9. The mass transfer assembly of claim 4, wherein the expansion medium is made of nitrogen, argon, water, or alcohol, and the flexible layer is made of polyester fiber or polyvinyl alcohol.

10. A display panel, comprising a driving substrate and a plurality of light-emitting elements disposed on one side of the driving substrate, wherein the plurality of light-emitting elements are transferred by a mass transfer assembly from a growth substrate to the driving substrate, each of the plurality of light-emitting elements comprises a light-emitting body initially connected with the growth substrate and two electrodes connected with the light-emitting body, and the mass transfer assembly comprises: a temporary substrate, wherein the temporary substrate is configured to adhere two electrodes of each of a plurality of light-emitting elements on the growth substrate, to separate the plurality of light-emitting elements from the growth substrate, such that a light-emitting body of each of the plurality of light-emitting elements is suspended; and a transfer substrate having a plurality of adhesion regions, wherein the transfer substrate is configured to adhere, through the adhesion regions, a light-emitting body of each of a plurality of light-emitting elements on the temporary substrate, to separate the plurality of light-emitting elements from the temporary substrate, such that two electrodes of each of the plurality of light-emitting elements are suspended; wherein one adhesion region adheres a light-emitting body of one light-emitting element; wherein positions of the plurality of adhesion regions on the transfer substrate are in one-to-one correspondence with positions of a plurality of binding regions on the driving substrate; wherein the transfer substrate is further configured to transfer a plurality of light-emitting elements on the transfer substrate to the side of the driving substrate, to bind two electrodes of each of the plurality of light-emitting elements to a corresponding binding region of the driving substrate.

11. The display panel of claim 10, wherein the temporary substrate comprises a first base substrate and an adhesion layer, the adhesion layer is disposed on a surface of the first base substrate, and the adhesion layer is used for adhering the two electrodes of each of the plurality of light-emitting

elements on the growth substrate.

12. The display panel of claim 10, wherein the transfer substrate comprises a second substrate, a plurality of photothermal elements, a packaging layer, a plurality of expansion media, a flexible layer, and a plurality of adhesion elements, wherein the plurality of photothermal elements are spaced apart from one another on a surface of the second substrate; the packaging layer covers the plurality of photothermal elements on the second substrate, and the packaging layer defines a plurality of accommodating holes penetrating through the packaging layer; each of the plurality of accommodating holes is filled with an expansion medium, and the expansion medium is in contact with the photothermal element; the flexible layer shields and covers the plurality of expansion media, the plurality of adhesion elements are disposed on a surface of the flexible layer facing away from the plurality of expansion media, and a position of one expansion medium and a position of one adhesion element both correspond to a position of one adhesion region; the photothermal element is configured to receive lights to generate heat; and the expansion medium is configured to receive heat conducted by the photothermal element to expand to abut against the flexible layer, and the flexible layer abuts against the adhesion element, such that the adhesion element adheres to the light-emitting body of the light-emitting element.

13. The display panel of claim 10, wherein the transfer substrate comprises a flat layer, a plurality of electric heating elements, a packaging layer, a plurality of expansion media, a plurality of flexible layers, and a plurality of adhesion elements, wherein the plurality of electric heating elements are disposed on a surface of the flat layer; the packaging layer covers the plurality of electric heating elements on the flat layer, and the packaging layer defines a plurality of accommodating holes penetrating through the packaging layer; each of the plurality of accommodating holes is filled with the expansion medium, and the expansion medium is in contact with the electric heating element; the flexible layer covers the plurality of expansion media, the plurality of adhesion elements are disposed on a surface of the flexible layer facing away from the plurality of expansion media, and a position of one expansion medium and a position of one adhesion element both correspond to a position of one adhesion region; the electric heating element is configured to receive a current to generate heat; and the expansion medium is configured to receive heat conducted by the electric heating element to expand to abut against the flexible layer, and the flexible layer abuts against the adhesion element, such that the adhesion element adheres to the light-emitting body of the light-emitting element.

14. The display panel of claim 13, wherein the transfer substrate further comprises a plurality of control transistors, wherein each of the plurality of control transistors has a gate, an active device, a drain, and a source, and the active device is electrically connected with the drain and the source; the drain of the control transistor is further electrically connected with one electric heating element; the source of the control transistor is further electrically connected with a power-supply terminal; and the gate of the control transistor is configured to receive a control signal to turn on the active device of the control transistor, to make the drain of the control transistor be electrically connected with the source of the control transistor, such that the electric heating element receives a current outputted by the power-supply terminal.

15. The display panel of claim 14, wherein the transfer substrate further comprises a second substrate, an insulating layer, and a passivation layer, wherein the second substrate is disposed on one side of the flat layer facing away from the plurality of electric heating elements and is spaced apart from the flat layer; gates of the plurality of control transistors are spaced apart from one another on one side of the second substrate facing the flat layer; the insulating layer covers the gates of the plurality of control transistors on the second substrate; active devices of the plurality of control transistors are disposed on a surface of the insulating layer facing away from the second substrate, a position of one active device corresponds to a position of one gate, the drain and the source are connected with the active device, and the drain is spaced apart from the source; and the passivation layer covers the active devices, drains, and sources of the plurality of control transistors

on the insulating layer, and the passivation layer is connected with the flat layer.

16. The display panel of claim 15, wherein the flat layer defines a plurality of first vias penetrating through the flat layer, the passivation layer defines a plurality of second vias penetrating through the passivation layer, one drain is exposed from one second via, and one second via communicates with one first via; and the transfer substrate further comprises a plurality of connecting electrodes, one connecting electrode is accommodated in one first via and one second via which are in communication, and one connecting electrode is connected with one electric heating element and one drain.

17. The display panel of claim 12, wherein the expansion medium is made of nitrogen, argon, water, or alcohol, and the flexible layer is made of polyester fiber or polyvinyl alcohol.

18. The display panel of claim 13, wherein wherein the expansion medium is made of nitrogen, argon, water, or alcohol, and the flexible layer is made of polyester fiber or polyvinyl alcohol.

19. A display device, comprising a housing and a display panel, wherein the display panel is received in the housing, a light-emitting side of the display panel is exposed from the housing, and the display panel comprises a driving substrate and a plurality of light-emitting elements disposed on one side of the driving substrate; wherein the plurality of light-emitting elements are transferred by a mass transfer assembly from a growth substrate to the driving substrate, each of the plurality of light-emitting elements comprises a light-emitting body initially connected with the growth substrate and two electrodes connected with the light-emitting body, and the mass transfer assembly comprises: a temporary substrate, wherein the temporary substrate is configured to adhere two electrodes of each of a plurality of light-emitting elements on the growth substrate, to separate the plurality of light-emitting elements from the growth substrate, such that a light-emitting body of each of the plurality of light-emitting elements is suspended; and a transfer substrate having a plurality of adhesion regions, wherein the transfer substrate is configured to adhere, through the adhesion regions, a light-emitting body of each of a plurality of light-emitting elements on the temporary substrate, to separate the plurality of light-emitting elements from the temporary substrate, such that two electrodes of each of the plurality of light-emitting elements are suspended; wherein one adhesion region adheres a light-emitting body of one light-emitting element; wherein positions of the plurality of adhesion regions on the transfer substrate are in one-to-one correspondence with positions of a plurality of binding regions on the driving substrate; wherein the transfer substrate is further configured to transfer a plurality of light-emitting elements on the transfer substrate to the side of the driving substrate, to bind two electrodes of each of the plurality of light-emitting elements to a corresponding binding region of the driving substrate.

20. The display device of claim 19, wherein the transfer substrate comprises a second substrate, a plurality of photothermal elements, a packaging layer, a plurality of expansion media, a flexible layer, and a plurality of adhesion elements, wherein the plurality of photothermal elements are spaced apart from one another on a surface of the second substrate; the packaging layer covers the plurality of photothermal elements on the second substrate, and the packaging layer defines a plurality of accommodating holes penetrating through the packaging layer; each of the plurality of accommodating holes is filled with an expansion medium, and the expansion medium is in contact with the photothermal element; the flexible layer shields and covers the plurality of expansion media, the plurality of adhesion elements are disposed on a surface of the flexible layer facing away from the plurality of expansion media, and a position of one expansion medium and a position of one adhesion element both correspond to a position of one adhesion region; the photothermal element is configured to receive lights to generate heat; and the expansion medium is configured to receive heat conducted by the photothermal element to expand to abut against the flexible layer, and the flexible layer abuts against the adhesion element, such that the adhesion element adheres to the light-emitting body of the light-emitting element.
